Nuclear physics – the heart of the atom

The atomic nucleus is a hundred-thousandth of the diameter of the atom, yet it contains almost all of its mass and a million times more energy than the chemical bonds. Nuclear physics explains why the Sun shines, why uranium can be split and iron cannot, why some nuclei live for billions of years and others for microseconds – and how the elements we are made of were created in stars. This page covers nuclear physics from Rutherford to the present day with three interactive figures.

The discovery of the nucleus

In 1909 Hans Geiger and Ernest Marsden fired alpha particles at a thin gold foil in Ernest Rutherford's laboratory in Manchester. Most went straight through, but about one in 8000 was thrown back. Rutherford compared it to firing a shell at a piece of tissue paper and having it come back and hit you. In 1911 he showed that the result can only be explained if almost all the mass and positive charge of the atom is concentrated in a small nucleus. Two years later Niels Bohr built his model of the atom on Rutherford's nucleus.

In 1919 Rutherford fired alpha particles into nitrogen and got hydrogen nuclei out – the first artificial transmutation of an element. He called the hydrogen nucleus the *proton*. But nuclei weighed about twice as much as the protons could account for. In 1932 James Chadwick found the neutron, and the picture was complete: the nucleus consists of Z protons and N neutrons, $A = Z + N$ nucleons in all.

Size and density

Electron-scattering experiments (Robert Hofstadter, 1950s) showed that the radius of the nucleus grows as the cube root of the number of nucleons:

$$R \approx r_0 \, A^{1/3}, \qquad r_0 \approx 1.2 \ \text{fm} = 1.2 \cdot 10^{-15} \ \text{m}$$

The volume is thus proportional to the number of nucleons: the nucleons are packed like drops in a liquid of constant density – about $2.3 \cdot 10^{17}$ kg/m³. A teaspoon of nuclear matter would weigh about a billion tonnes. In a neutron star the whole star has that density.

The nuclear force

The protons in a nucleus repel each other electrically with forces of several tens of newtons – enormous for such small particles. Something must hold them together. The strong nuclear force is about a hundred times stronger than the electric force at short distances, but it reaches only 1–2 femtometres. It is almost the same between proton–proton, proton–neutron and neutron–neutron, which Heisenberg expressed with isospin (see Isospin).

In 1935 Hideki Yukawa proposed that the force is carried by a new particle. Because the force is short-ranged, the particle must have mass: $mc^2 \approx \hbar c / r \approx 200\ \text{MeV}\cdot\text{fm} / 1.4\ \text{fm} \approx 140$ MeV. The pion was found in cosmic rays in 1947 with exactly that mass. Today we know that the nuclear force is a residue of the strong interaction between quarks – just as the van der Waals forces between molecules are a residue of electromagnetism.

Binding energy

A nucleus weighs less than the protons and neutrons it is made of. The difference, the mass defect, corresponds through Einstein's $E = mc^2$ to the energy needed to take the nucleus apart. For helium-4 it is 28.3 MeV – almost 1 % of the mass. In the 1920s Francis Aston measured these small mass differences with his mass spectrograph and gave Arthur Eddington the idea that the Sun gets its energy by converting hydrogen into helium.

In 1935 Carl Friedrich von Weizsäcker set up a formula for the binding energy based on the picture of the nucleus as a charged liquid drop – *the semi-empirical mass formula*:

$$B(Z,A) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} \pm \delta(A)$$

Volume term

Each nucleon is bound to its neighbours: $a_V \approx 15.8$ MeV per nucleon.

Surface term

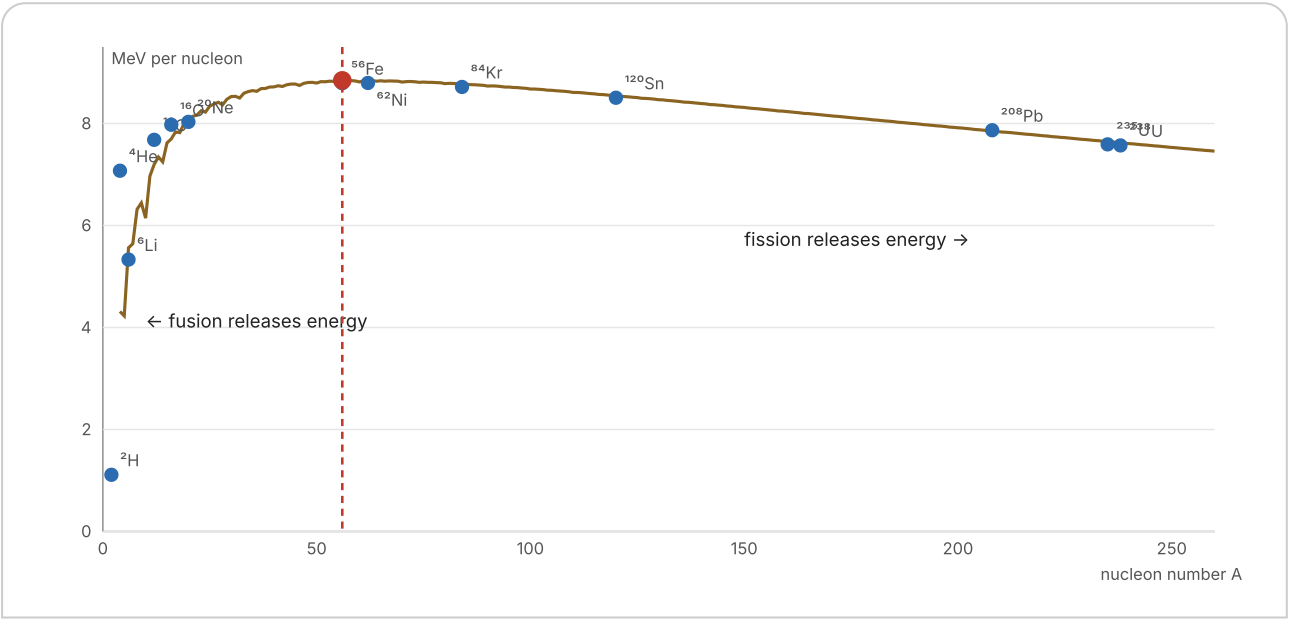
Nucleons at the surface have fewer neighbours – like surface tension in a drop.

Coulomb term

The electric repulsion of the protons. Grows as Z^2 and makes heavy nuclei unstable.

Asymmetry and pairing terms

Quantum mechanics: the nucleus prefers equal numbers of protons and neutrons (the Pauli principle), and even numbers of each (pairing).



Shells and magic numbers

The liquid-drop model cannot explain why nuclei with 2, 8, 20, 28, 50, 82 or 126 protons or neutrons are especially stable. Tin has ten stable isotopes (Z = 50), and lead-208 with 82 protons and 126 neutrons is “doubly magic”. Like the electrons in an atom, the nucleons arrange themselves in shells. But a simple shell model gave the wrong numbers. In 1949 Maria Goeppert Mayer, and independently Hans Jensen, realised that the spin of the nucleons couples strongly to their orbital motion (spin–orbit coupling). This splits the levels in exactly the way that produces the magic numbers. They received the Nobel Prize in 1963.

The collective model – from Copenhagen

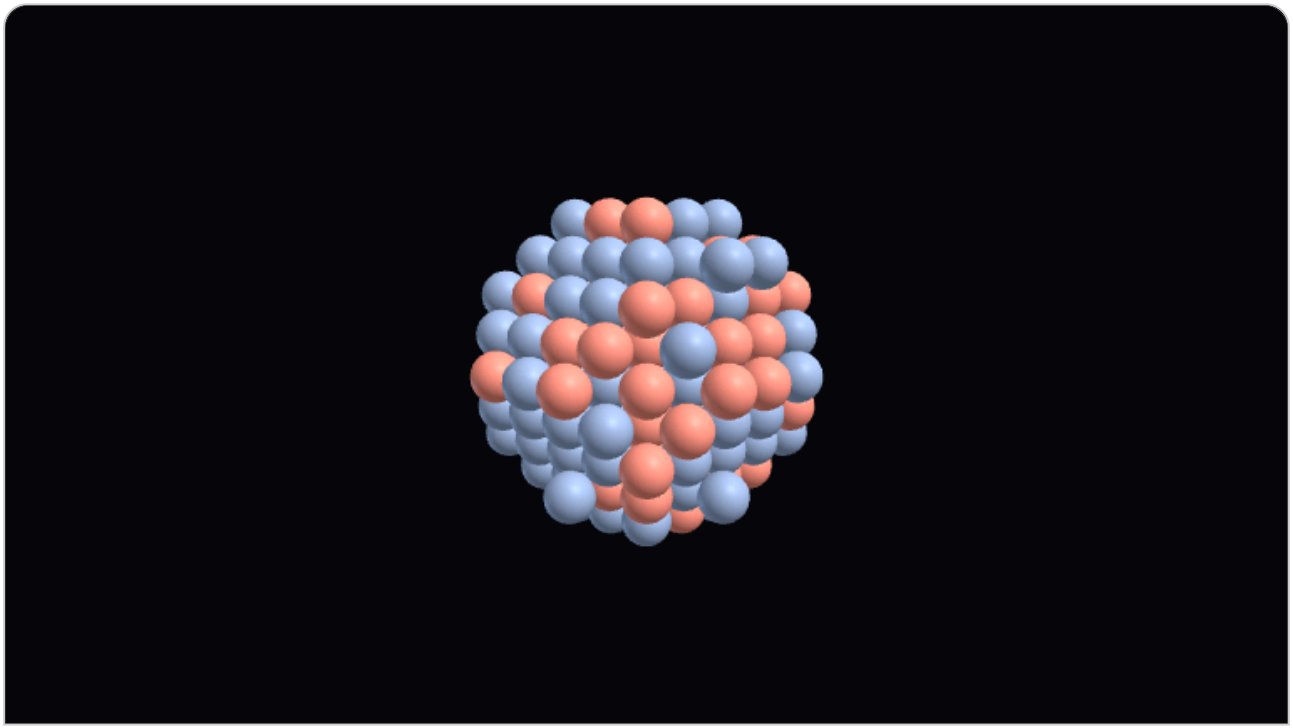
The shell model treats the nucleons individually, the liquid-drop model the nucleus as a whole. Many nuclei are not spherical but elongated like a rugby ball, and they can rotate and vibrate. Niels Bohr’s son Aage Bohr and the American Ben Mottelson united the two pictures at the Niels

Bohr Institute in the 1950s: the nucleons move in a field whose shape itself changes. Deformed nuclei show rotational bands with energies

$$E_I = \frac{\hbar^2}{2\mathcal{I}}\, I(I+1), \qquad I = 0, 2, 4, \ldots$$

exactly like a quantum-mechanical spinning top. Aage Bohr, Mottelson and James Rainwater received the Nobel Prize in 1975. As early as 1936 Niels Bohr had proposed the compound nucleus: a neutron striking a nucleus quickly shares its energy with all the nucleons, and the nucleus "forgets" how it was formed. That picture became the basis for understanding fission.

The nucleus in 3D



Radioactive decay

Alpha (α)

The nucleus emits a helium nucleus: $Z \to Z-2$, $A \to A-4$. Typical of heavy nuclei. Alpha particles are stopped by a sheet of paper.

Beta (β^- and β^+)

A neutron turns into a proton, an electron and an antineutrino (or the reverse with a positron and a neutrino). Mediated by the weak force. The neutrino was predicted by Pauli and the theory written by Fermi.

Gamma (γ)

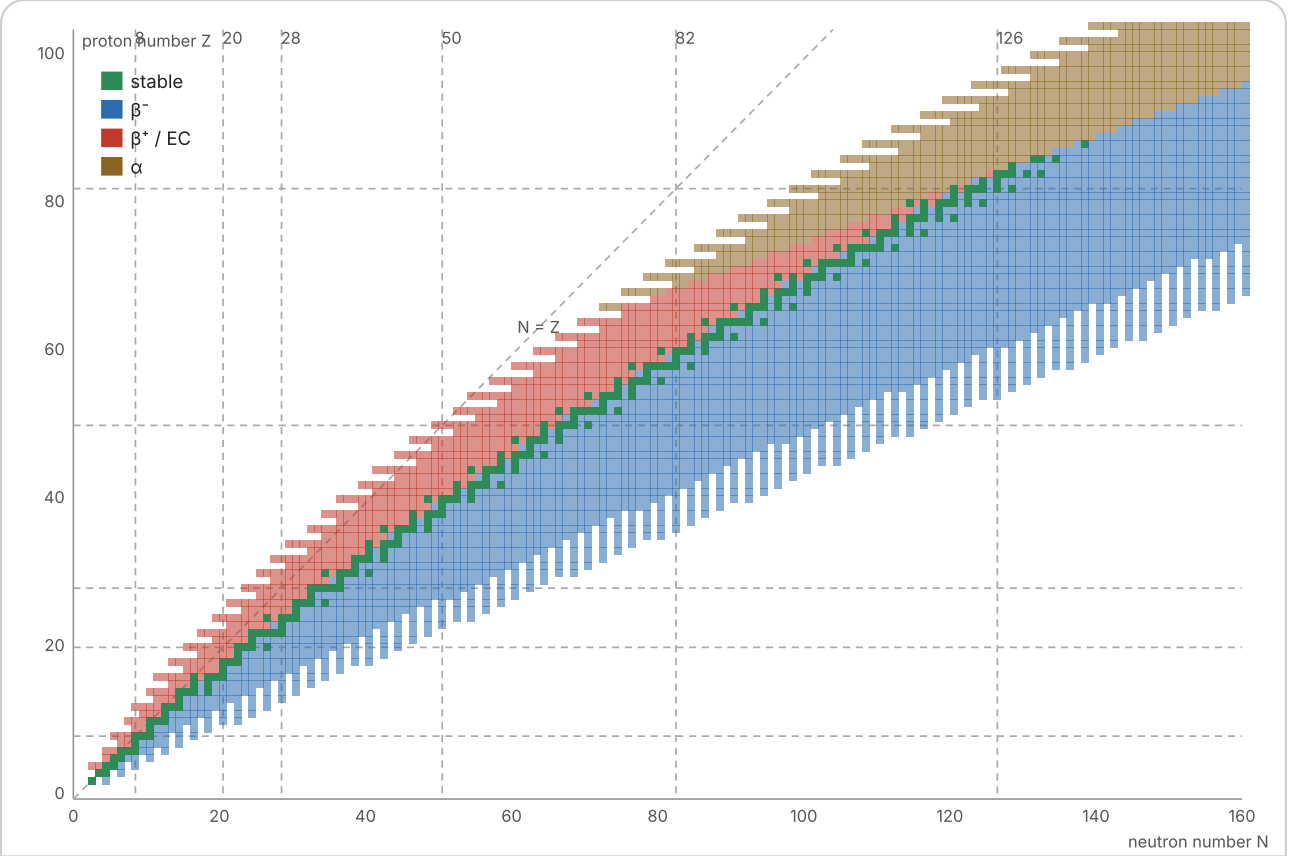
An excited nucleus drops to a lower level and emits a photon – just like an atom, but with a million times more energy.

The decay law

Each nucleus has a fixed probability per unit time of decaying: $N(t) = N_0 e^{-\lambda t}$, with half-life $T_{1/2} = \ln 2 / \lambda$. See Marie Curie.

The chart of nuclides

Plot all nuclei in a diagram with neutron number horizontally and proton number vertically, and the roughly 250 stable nuclei lie in a narrow band – the valley of stability. Light nuclei have $N \approx Z$, heavy nuclei have more neutrons, because the extra neutrons dilute the repulsion of the protons. Nuclei above the valley have too many protons and decay by β^+ or electron capture; nuclei below the valley decay by β^-. The heaviest decay by α decay or spontaneous fission.

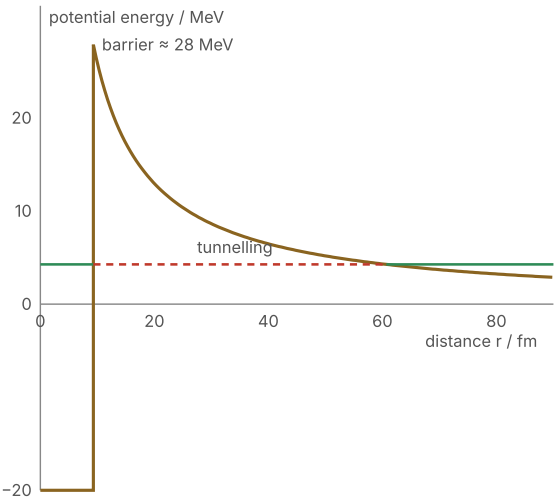


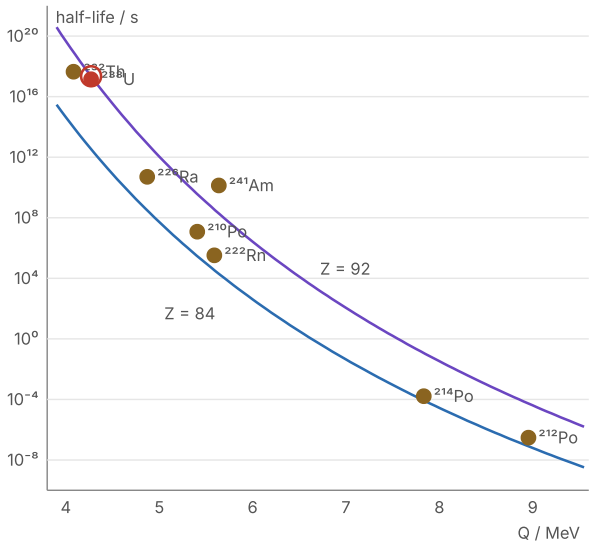
Alpha decay and tunnelling

Alpha decay was a puzzle. Uranium-238 emits alpha particles with 4.3 MeV, but the Coulomb barrier around the nucleus is about 30 MeV high. Classically the particle could never escape. And the half-lives are bizarre: as early as 1911 Geiger and Nuttall found that a small change in energy gives a huge change in lifetime. Polonium-212 emits alpha particles with 9 MeV and has a half-life of 0.3 microseconds; uranium-238 with 4.3 MeV lives for 4.5 billion years – a factor of 10^{23}.

In 1928 George Gamow, then in Göttingen and shortly afterwards in Copenhagen, and independently Ronald Gurney and Edward Condon, explained that the alpha particle *tunnels* through the barrier – a purely quantum-mechanical effect. The probability of getting through depends exponentially on the width of the barrier, and the width depends on the energy. It was the first application of quantum mechanics to the nucleus.

$$P \approx e^{-2G}, \qquad G \approx \frac{\pi Z_1 Z_2 e^2}{4\pi\varepsilon_0 \hbar v} \quad \Rightarrow \quad \log_{10} T_{1/2} \approx \frac{aZ + b}{\sqrt{Q}} + cZ + d$$





Fission

In December 1938 Otto Hahn and Fritz Strassmann found barium in uranium that had been irradiated with neutrons, and Lise Meitner and Otto Frisch explained that the nucleus had split. In 1939 Niels Bohr and John Wheeler worked out the theory from the liquid-drop model: the neutron gives the nucleus enough energy to start oscillating, it develops a waist, and the Coulomb repulsion tears it apart. They predicted that it is the rare isotope uranium-235 that is split by slow neutrons. Each fission releases about 200 MeV and two or three new neutrons, so a chain reaction is possible. Fermi started the first reactor in Chicago in 1942. See also The physics of nuclear weapons.

Nature got there first, though. In 1972 French scientists found traces in the uranium mine at Oklo in Gabon showing that about two billion years ago the uranium had worked as a natural reactor – at that time the proportion of uranium-235 was high enough, and groundwater acted as moderator.

Fusion and the stars

Light nuclei can release energy by fusing. But they must come so close that the nuclear force can act, and that requires them to overcome the Coulomb barrier. Even at the centre of the Sun, at 15 million degrees, the typical energy of the particles is only about 1 keV, far below the barrier. Once again it is tunnelling that makes it possible. In 1938–39 Hans Bethe showed how the Sun converts hydrogen into helium through the pp chain, and heavier stars through the CNO cycle, and received the Nobel Prize in 1967:

$$4\,{}^{1}\mathrm{H} \;\rightarrow\; {}^{4}\mathrm{He} + 2e^{+} + 2\nu_e + 26.7\ \mathrm{MeV}$$

In 1957 Margaret and Geoffrey Burbidge, William Fowler and Fred Hoyle described how all the elements up to iron are formed by fusion in stars, and the heavier ones by neutron capture – slowly in red giants (the s-process) and rapidly in explosive events (the r-process). In 2017, with the gravitational wave GW170817, two neutron stars were seen to merge, and the light showed that heavy elements were formed in a kilonova. The gold in a wedding ring was probably made like that.

On Earth fusion is hard to control. In December 2022 the National Ignition Facility in California for the first time obtained more fusion energy (3.15 MJ) than the laser energy (2.05 MJ) that struck the fuel capsule. The international tokamak ITER is being built in France.

Applications

Carbon-14 dating

Cosmic rays produce carbon-14 in the atmosphere, and living organisms take it up. When they die, it decays with a half-life of 5730 years. Willard Libby developed the method in 1949 (Nobel Prize in Chemistry 1960).

Tracers

The Hungarian George de Hevesy introduced radioactive tracers into biology, partly at the Niels Bohr Institute, and received the Nobel Prize in Chemistry in 1943. When the Germans occupied Denmark, he dissolved von Laue's and Franck's Nobel medals in aqua regia so that they could not be seized. The gold was later recovered and the medals recast.

Medicine

PET scanning uses positron-emitting isotopes such as fluorine-18, radiotherapy uses gamma rays and protons, and MRI scanning exploits the spin of hydrogen nuclei.

Energy

Nuclear power supplies about 9 % of the world's electricity. In 1985 Denmark decided not to build nuclear power stations, but the research reactors at Risø operated from 1957 to 2001.

Further reading

- Kenneth S. Krane: *Introductory Nuclear Physics* (Wiley 1988)
- Frank Close: *Radioactivity. A Very Short Introduction* (Oxford 2011)
- Richard Rhodes: *The Making of the Atomic Bomb* (1986)
- Aage Bohr and Ben R. Mottelson: *Nuclear Structure* I–II (1969–75)
- The biographies of Rutherford, Curie, Meitner, Fermi and Bohr

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